Financial Statement Analysis

Ratio Analysis Problems: Britannia Industries FY2026

Six problems, one for each ratio family, based on the standalone financial statements of Britannia Industries Limited for the year ended 31 March 2026

Problem edition: problems and formulas

How the problems are organised

Each problem has a short narrative, a data table, a formulas-and-definitions section, and questions that name the ratio family and its purpose. Every formula is shown with its components defined.

No.	Ratio family	Ratios covered
1	Liquidity	Working capital; current ratio; quick ratio; broad quick ratio
2	Solvency	Total liabilities to total assets; total liabilities to equity; long-term debt to capitalization; times interest earned; cash-basis times interest earned; times fixed charges earned; preferred-stock-dividend coverage
3	Efficiency	Receivables to sales; collection period; inventory turnover; days in inventory; days' payables; total asset turnover; fixed asset turnover; capital turnover
4	Profitability	Gross, EBIT, pretax and net margins; return on assets; pretax operating return on assets; return on equity; return on total capital; return on tangible net worth
5	Market	Earnings per share; price-earnings ratio; capitalization rate; price to book value; dividend yield; dividend payout
6	Frameworks	Du Pont analysis; tax retention; extended Du Pont; Du Pont refinements; linking ratio; sustainable growth rate; Altman Z-score

Basis and conventions used throughout

- **Source and units.** Britannia standalone statements for the year ended 31 March 2026, with 31 March 2025 comparatives. Amounts are in ₹ crore, computed at full precision and shown rounded.
- **Net sales** = revenue from operations. **Cost of goods sold (COGS)** = cost of materials consumed + purchases of stock-in-trade + changes in inventories + conversion charges paid to contract packers. Conversion charges are included because Britannia outsources some production.
- **EBIT** = profit before tax + finance costs, used here as the measure of operating profit. It therefore includes other income.

- **Average balance** = (balance at 31 March 2025 + balance at 31 March 2026) ÷ 2. Ratios that need an average are solved for FY2026 only. Ratios that use year-end balances are solved for both years. A year has 365 days.
- **Share price.** A market price is needed in Problems 5 and 6. Use the **assumed price of ₹5,400 per share at 31 March 2026**, which is supplied for this exercise.

Problem 1: Liquidity Ratios

Ratio family: Liquidity. **Ratios covered:** working capital, current ratio, quick (acid-test) ratio.

Narrative

Britannia sells biscuits and dairy products to distributors on credit and buys its materials on supplier credit. It also keeps a large portfolio of investments inside current assets. A bank is considering a fresh working-capital limit and wants to know how comfortably the company can meet the obligations falling due within twelve months, and how much of that cover comes from cash-like assets rather than from inventories.

Data

Table 1.1: Britannia, selected balance sheet items at 31 March (₹ crore)

Item	FY2026	FY2025
Total current assets	4,187.68	3,436.43
Total current liabilities	3,940.22	3,348.13
Inventories	1,219.92	1,128.90
Current investments	1,617.02	1,071.12
Trade receivables (net)	404.74	379.63
Cash and cash equivalents	20.79	38.19
Other bank balances	70.03	67.45
Current loans (inter-corporate deposits)	121.00	50.00
Other current financial assets (mainly state-incentive receivables)	636.69	609.46

Formulas and definitions

Use these definitions to answer the questions below.

Question (a): Working capital and current ratio

> **Working capital** = Current assets − Current liabilities
>
> **Current ratio** = Current assets ÷ Current liabilities

where:

- **Current assets:** assets expected to turn into cash within twelve months (cash, investments, receivables, inventories and similar).
- **Current liabilities:** obligations falling due within twelve months (trade payables, short-term borrowings, current maturities of term loans, provisions and similar).

Question (b): Quick ratio

> **Quick ratio =** Quick assets ÷ Current liabilities
>
> **Quick assets =** Cash and cash equivalents + Other bank balances + Current investments + Trade receivables

where:

- **Quick assets:** current assets that can be turned into cash quickly at close to book value. Inventories are excluded because they must first be sold.
- **Current liabilities:** obligations falling due within twelve months.

Question (c): Broad quick ratio

> **Broad quick ratio =** (Quick assets + Current loans + Other current financial assets) ÷ Current liabilities

where:

- **Current loans:** short-term inter-corporate deposits placed with other companies, repayable within a year.
- **Other current financial assets:** security deposits, accrued interest and incentives recoverable from governments under industrial-policy schemes.

Questions

(a) Liquidity ratios. *Purpose: to measure the size of the short-term cushion.* Compute working capital and the current ratio for both years.

(b) Liquidity ratios. *Purpose: to test the ability to pay current liabilities immediately from near-cash assets.* Compute the quick ratio for both years, counting cash, other bank balances, current investments and trade receivables as quick assets.

(c) Liquidity ratios. *Purpose: to judge whether the quick ratio understates Britannia's real liquidity.* Recompute the quick ratio after adding current loans and other current financial assets to quick assets, and comment on the trend.

Problem 2: Solvency Ratios

Ratio family: Solvency. **Ratios covered:** total liabilities to total assets, total liabilities to equity, long-term debt to capitalization, times interest earned, cash-basis times interest earned, times fixed charges earned, preferred-stock-dividend coverage.

Narrative

In FY2025 Britannia redeemed ₹690.56 crore of bonus debentures, and in FY2026 it repaid more term loans while drawing on working-capital loans. A bond-rating analyst wants to know how heavily creditors finance the company, and whether earnings comfortably cover interest, lease rentals and any preferred dividends. Britannia's rentals are short-term leases charged to other expenses, and it has no preferred shares.

Data

Table 2.1: Britannia, items needed for solvency analysis (₹ crore)

Item	FY2026	FY2025
Profit before tax	3,293.34	2,867.77
Finance costs (interest)	110.80	137.10
Depreciation and amortisation	305.34	284.67
Lease rentals (short-term leases)	77.63	90.28
Tax expense	731.62	737.05
Preferred stock dividends	Nil	Nil
Total liabilities	4,299.21	4,133.17
Total assets	8,940.44	8,019.72
Total equity	4,641.23	3,886.55
Non-current borrowings	285.83	712.50
Non-current lease liabilities	0.36	0.48
Current maturities of long-term bank debt	426.67	191.67

Formulas and definitions

Use these definitions to answer the questions below.

Question (a): Capital-structure ratios

> **Total liabilities to total assets** = Total liabilities ÷ Total assets
>
> **Total liabilities to equity** = Total liabilities ÷ Total equity
>
> **Long-term debt to capitalization** = Long-term debt ÷ (Long-term debt + Total equity)

where:

- **Total liabilities:** all current and non-current liabilities.

- **Long-term debt:** non-current borrowings + non-current lease liabilities + current maturities of long-term bank debt. Working-capital loans are not included.
- **Capitalization:** long-term debt + total equity, that is, the permanent funds of the business.

Question (b): Interest coverage

> **EBIT =** Profit before tax + Finance costs
>
> **Times interest earned =** EBIT ÷ Finance costs
>
> **Cash-basis times interest earned =** (EBIT + Depreciation and amortisation) ÷ Finance costs

where:

- **EBIT:** operating profit before interest. It includes other income.
- **Finance costs:** interest on borrowings and lease liabilities, plus other finance charges.
- **Depreciation and amortisation:** the non-cash charge added back to show cover from funds generated by operations.

Question (c): Fixed-charge and preferred-dividend coverage

> **Times fixed charges earned =** (EBIT + Lease rentals) ÷ (Finance costs + Lease rentals + Pretax equivalent of preferred dividends)
>
> **Pretax equivalent of preferred dividends =** Preferred dividends ÷ (1 − Tax rate)
>
> **Tax rate =** Tax expense ÷ Profit before tax
>
> **Preferred-stock-dividend coverage =** Net income ÷ Preferred dividends

where:

- **Lease rentals:** rent paid under leases, which is a fixed charge. The numerator adds it back because it is deducted before EBIT. Only rentals expensed in other expenses are used; the ₹0.07 crore lease interest is already in finance costs.
- **Pretax equivalent of preferred dividends:** preferred dividends are paid out of after-tax profit, so they are grossed up to the pre-tax profit needed to pay them.
- **Preferred-stock-dividend coverage:** calculated after tax, because preferred dividends are not tax-deductible.

Questions

(a) Solvency ratios (capital structure). *Purpose: to see how much of the assets and of the long-term capital is financed by creditors.* Compute total liabilities to total assets, total liabilities to equity and long-term debt to capitalization for both years.

(b) Solvency ratios (earnings coverage). *Purpose: to test whether operating profit comfortably covers interest, first on a pretax basis and then on a cash basis.* Compute times interest earned and the cash-basis times interest earned for both years.

(c) Solvency ratios (fixed-charge coverage). *Purpose: to test cover for all fixed charges, including lease rentals and preferred dividends.* Compute times fixed charges earned, with and without depreciation added back, and the preferred-stock-dividend coverage, for both years.

Problem 3: Efficiency Ratios

Ratio family: Efficiency (activity). **Ratios covered:** receivables to sales, collection period, inventory turnover, days in inventory, days' payables, total asset turnover, fixed asset turnover, capital turnover.

Narrative

Britannia's management reviews how quickly the business converts working capital into cash and how hard its assets work. Britannia collects from distributors, holds raw materials and finished stock, and pays suppliers. The analyst has the FY2026 income statement items and the opening and closing balances, and wants to measure the working-capital cycle and the productivity of assets and capital.

Data

Table 3.1: Britannia, balances at 31 March (₹ crore)

Item	FY2026	FY2025
Trade receivables (net)	404.74	379.63
Inventories	1,219.92	1,128.90
Trade payables	1,781.87	1,592.40
Total assets	8,940.44	8,019.72
Property, plant and equipment + capital work-in-progress	2,464.72	2,591.38
Long-term debt (incl. current maturities and leases)	712.86	904.65
Total equity	4,641.23	3,886.55

Table 3.2: Britannia, FY2026 income statement items (₹ crore)

Item	FY2026
Revenue from operations (net sales)	18,445.82
Cost of materials consumed	9,045.68
Purchases of stock-in-trade	2,028.01
Changes in inventories of finished goods, WIP and stock-in-trade	(21.42)
Conversion charges (contract packers)	764.69
Purchases of raw and packing materials, net (disclosed in Note 28)	9,152.52

Formulas and definitions

Use these definitions to answer the questions below.

Question (a): Receivables

> **Receivables to sales =** Average trade receivables ÷ Net sales
>
> **Average collection period (days) =** Receivables to sales × 365

where:

- **Average trade receivables:** (receivables at 31 March 2025 + at 31 March 2026) ÷ 2, net of loss allowance.
- **Net sales:** revenue from operations.

Question (b): Inventory

> **Cost of goods sold =** Cost of materials consumed + Purchases of stock-in-trade + Changes in inventories + Conversion charges
>
> **Inventory turnover =** Cost of goods sold ÷ Average inventories
>
> **Days in inventory =** 365 ÷ Inventory turnover

where:

- **Changes in inventories:** a negative figure means finished stock increased, which reduces the cost charged to the year.
- **Average inventories:** (inventories at 31 March 2025 + at 31 March 2026) ÷ 2.

Question (c): Payables and the working-capital cycle

> **Purchases =** Purchases of raw and packing materials + Purchases of stock-in-trade
>
> **Payables to purchases =** Year-end trade payables ÷ Purchases
>
> **Days' payables =** Payables to purchases × 365
>
> **Net working-capital cycle (days) =** Collection period + Days in inventory − Days' payables

where:

- **Purchases:** the actual purchases disclosed in the notes to the accounts. They are a minimum figure, because services such as freight and conversion charges are excluded.
- **Year-end trade payables:** the balance at 31 March 2026.

Question (d): Asset and capital turnover

> **Total asset turnover =** Net sales ÷ Average total assets
>
> **Fixed asset turnover =** Net sales ÷ Average (Property, plant and equipment + Capital work-in-progress)
>
> **Capital turnover =** Net sales ÷ Average total capital

where:

- **Total capital:** long-term debt + total equity, the permanent funds used in the business.
- **Averages:** each is (31 March 2025 balance + 31 March 2026 balance) ÷ 2.

Questions

(a) Efficiency ratios (receivables). *Purpose: to judge credit policy and collection speed.* Compute receivables to sales and the average collection period for FY2026.

(b) Efficiency ratios (inventory). *Purpose: to judge how quickly stock is converted into sales.* Compute the inventory turnover and the days in inventory for FY2026.

(c) Efficiency ratios (payables). *Purpose: to see how much supplier credit funds the business.* Compute payables to purchases and the days' payables for FY2026, then combine (a) to (c) into a net working-capital cycle.

(d) Efficiency ratios (asset and capital use). *Purpose: to measure the sales generated per rupee invested.* Compute total asset turnover, fixed asset turnover and capital turnover for FY2026.

Problem 4: Profitability Ratios

Ratio family: Profitability. **Ratios covered:** gross, EBIT, pretax and net margins; return on total assets; pretax operating return on assets; return on equity; return on total capital; return on tangible net worth.

Narrative

Britannia's net profit grew from ₹2,130.72 crore to ₹2,561.72 crore in FY2026. An equity analyst wants to know whether the improvement came from better margins on each rupee of sales or from higher returns on the money invested, and how the answer changes when profit is measured against total assets, long-term capital and shareholders' funds.

Data

Table 4.1: Britannia, FY2026 income statement items (₹ crore)

Item	FY2026
Revenue from operations (net sales)	18,445.82
Cost of goods sold (as defined on page 2)	11,816.96
Finance costs	110.80
Profit before tax	3,293.34
Net income (profit for the year)	2,561.72

Table 4.2: Britannia, balances at 31 March (₹ crore)

Item	FY2026	FY2025
Total assets	8,940.44	8,019.72
Total equity	4,641.23	3,886.55
Total capital (long-term debt + equity)	5,354.09	4,791.20
Intangible assets (software, trademarks; no goodwill)	3.39	5.96

Formulas and definitions

Use these definitions to answer the questions below.

Question (a): Margins

> **Gross margin =** (Net sales − Cost of goods sold) ÷ Net sales
>
> **EBIT margin =** EBIT ÷ Net sales
>
> **Pretax margin =** Profit before tax ÷ Net sales
>
> **Net margin =** Net income ÷ Net sales

where:

- **EBIT:** profit before tax + finance costs.

- **Net income:** profit for the year after tax.

Question (b): Returns on investment

> **Pretax operating return on total assets =** EBIT ÷ Average total assets
>
> **Return on total assets =** Net income ÷ Average total assets
>
> **Return on total capital =** Net income ÷ Average total capital
>
> **Return on equity =** Net income ÷ Average total equity

where:

- **Average total assets:** (total assets at 31 March 2025 + at 31 March 2026) ÷ 2.
- **Average total capital:** (long-term debt + equity at 31 March 2025, plus the same at 31 March 2026) ÷ 2.
- **Average total equity:** (equity at 31 March 2025 + at 31 March 2026) ÷ 2.
- The first ratio is before tax and before interest, so it is not influenced by how assets are financed. The other three are after tax and interest.

Question (c): Return on tangible net worth

> **Return on tangible net worth =** Net income ÷ (Average total equity − Average intangible assets)

where:

- **Intangible assets:** tangible net worth is equity less intangible assets such as goodwill. Britannia has no goodwill, so its intangible assets (software and trademarks) are deducted.
- **Average intangible assets:** (intangibles at 31 March 2025 + at 31 March 2026) ÷ 2.

Questions

(a) Profitability ratios (margins). *Purpose: to see how much of each rupee of sales is kept at each level of profit.* Compute the gross margin, EBIT margin, pretax margin and net margin for FY2026.

(b) Profitability ratios (return on investment). *Purpose: to compare profit with the funds used under three views of investment: total assets, long-term capital and shareholders' funds.* Compute the pretax operating return on total assets, the return on total assets, the return on total capital and the return on equity for FY2026.

(c) Profitability ratios (conservative return). *Purpose: to measure the return on shareholders' funds after removing intangible assets.* Compute the return on tangible net worth for FY2026 and compare it with the return on equity.

Problem 5: Market Ratios

Ratio family: Market. **Ratios covered:** earnings per share, price-earnings ratio, capitalization rate, price to book value, dividend yield, dividend payout.

Narrative

An investor is deciding whether to buy Britannia shares after the FY2026 results. The company has no preferred stock and no dilutive securities, so basic and diluted earnings per share are the same. The investor wants to know what she pays for each rupee of earnings and book value, and how much cash comes back as dividends. The statements give no market price, so the exercise assumes a price of ₹5,400 per share at 31 March 2026.

Data

Table 5.1: Britannia, FY2026 per-share and market data

Item	FY2026
Net income (₹ crore)	2,561.72
Preferred dividends (₹ crore)	Nil
Shares outstanding (crore)	24.0868
Reported basic and diluted EPS (₹)	106.35
Total equity (₹ crore)	4,641.23
Dividend paid per share in the year (₹)	75.00
Dividends declared and paid in the year (₹ crore)	1,806.51
Assumed market price per share at 31 March 2026 (₹)	5,400.00

Formulas and definitions

Use these definitions to answer the questions below.

Question (a): EPS, price-earnings and capitalization rate

> **Earnings per share =** (Net income − Preferred dividends) ÷ Weighted average number of shares
>
> **Price-earnings ratio =** Market price per share ÷ Earnings per share
>
> **Capitalization rate =** Earnings per share ÷ Market price per share

where:

- **Weighted average number of shares:** 24.0868 crore shares, unchanged through the year because no shares were issued.
- **Capitalization rate:** the earnings return investors accept on the price, the reciprocal of the price-earnings ratio.

Question (b): Price to book value

> **Book value per share =** Total equity ÷ Number of shares outstanding
>
> **Price to book value =** Market price per share ÷ Book value per share

where:

- **Total equity:** all equity belongs to common shareholders, as there is no preferred stock.
- **Number of shares outstanding:** 24.0868 crore.

Question (c): Dividend yield and payout

> **Dividend yield =** Cash dividend per share ÷ Market price per share
>
> **Dividend payout ratio =** Cash dividends ÷ Net income

where:

- **Cash dividend per share:** ₹75.00 declared and paid in FY2026. The final dividend of ₹90.50 proposed after year-end is not counted.
- **Cash dividends:** ₹1,806.51 crore paid in the year.

Questions

(a) Market ratios (earnings). *Purpose: to find the price paid for each rupee of earnings and the earnings return that investors accept.* Verify the reported EPS, then compute the price-earnings ratio and its reciprocal, the capitalization rate.

(b) Market ratios (book value). *Purpose: to compare the market price with the net assets behind each share.* Compute the book value per share and the price to book value ratio.

(c) Market ratios (shareholder payout). *Purpose: to measure the cash return received and the share of profit distributed.* Compute the dividend yield and the dividend payout ratio.

Problem 6: Framework Ratios

Ratio family: Frameworks (integrated analysis). **Ratios covered:** Du Pont analysis, tax retention ratio, extended Du Pont, Du Pont refinements, linking ratio, sustainable growth rate, Altman Z-score.

Narrative

The research head of a brokerage firm wants a single explanation of Britannia's return on equity, a view on how fast profit can grow without raising new equity, and a check on financial distress risk. The analyst will break ROE into its drivers, link the ratios, estimate sustainable growth from the dividend policy and apply the Altman Z-score model.

Data

Table 6.1: Britannia, FY2026 inputs (₹ crore unless stated)

Item	FY2026
Revenue from operations (net sales)	18,445.82
Profit before tax	3,293.34
Finance costs	110.80
Tax expense	731.62
Net income	2,561.72
Average total assets	8,480.08
Average total equity	4,263.89
Average total capital (long-term debt + equity)	5,072.64
Equity at 31 March 2025 (beginning of year)	3,886.55
Dividends paid; preferred dividends	1,806.51; nil
Current assets; current liabilities	4,187.68; 3,940.22
Retained earnings; total assets; total liabilities (31 March 2026)	4,367.77; 8,940.44; 4,299.21
Shares outstanding (crore); assumed market price (₹)	24.0868; 5,400.00

Formulas and definitions

Use these definitions to answer the questions below.

Question (a): Du Pont analysis of return on assets

> **Return on total assets** = Net margin × Total asset turnover
>
> **Net margin** = Net income ÷ Net sales
>
> **Total asset turnover** = Net sales ÷ Average total assets

Question (b): Extended Du Pont and tax retention

> **Tax retention ratio** = 1 − (Tax expense ÷ Profit before tax)
>
> **Total leverage** = Average total assets ÷ Average total equity
>
> **Return on equity** = Pretax margin × Asset turnover × Total leverage × Tax retention ratio

where:

- **Tax retention ratio:** the share of pretax profit kept after tax, equal to 1 minus the book tax rate.
- **Pretax margin:** profit before tax ÷ net sales.
- **Total leverage:** how many rupees of assets each rupee of equity supports. Average balances are used because the equation explains the return on average equity.

Question (c): Refinements and linking ratio

> **"Pure" operating margin** = (Profit before tax + Finance costs) ÷ Net sales
>
> **Adjustment ratio** = Profit before tax ÷ (Profit before tax + Finance costs)
>
> **Capital turnover** = Net sales ÷ Average total capital
>
> **Capital leverage** = Average total capital ÷ Average total equity
>
> **Capital-based ROE** = Pure margin × Adjustment ratio × Capital turnover × Capital leverage × Tax retention ratio
>
> **Return on total capital (linking)** = Capital turnover × Net margin

where:

- **Adjustment ratio:** converts the margin before interest back to the margin on reported pretax profit, so the chain still multiplies to ROE.
- **Total capital:** long-term debt + equity, averaged over the year.

Question (d): Sustainable growth

> **Sustainable growth rate** = ROE on beginning equity × (1 − Payout ratio)
>
> **ROE on beginning equity** = Net income ÷ Equity at the start of the year
>
> **Payout ratio** = (Common dividends + Preferred dividends) ÷ Net income

where:

- **Sustainable growth rate:** the maximum growth in earnings possible without issuing new equity, if the return on equity and payout stay unchanged.
- **Beginning equity:** equity at 31 March 2025, ₹3,886.55 crore.

Question (e): Altman Z-score

Z = 1.2 T1 + 1.4 T2 + 3.3 T3 + 0.6 T4 + 0.999 T5

where:

- **T1 = Working capital ÷ Total assets** (short-term liquidity relative to size).
- **T2 = Retained earnings ÷ Total assets** (accumulated profitability).
- **T3 = EBIT ÷ Total assets** (earning power of the assets).
- **T4 = Market value of equity ÷ Book value of total liabilities** (how far the equity cushion can fall before liabilities exceed assets).
- **T5 = Net sales ÷ Total assets** (asset turnover).
- All balances are at 31 March 2026. A Z-score above 2.99 is generally read as safe and below 1.80 as the distress zone.

Questions

(a) Framework ratios (Du Pont). *Purpose: to explain return on assets through margin and asset use.* Show that net margin × total asset turnover reproduces the return on total assets.

(b) Framework ratios (extended Du Pont). *Purpose: to explain return on equity through pretax margin, asset turnover, leverage and the tax retention ratio.* Compute the tax retention ratio and the four-link return on equity.

(c) Framework ratios (refinements and linking). *Purpose: to restate the analysis on long-term capital and before interest.* Compute the capital-based Du Pont with the "pure" operating margin and its adjustment ratio, and the linking ratio for return on total capital.

(d) Framework ratios (growth). *Purpose: to find the growth in earnings that can be financed internally.* Compute the sustainable growth rate, with preferred dividends included in the payout.

(e) Framework ratios (distress prediction). *Purpose: to test the probability of bankruptcy.* Compute the Altman Z-score at 31 March 2026 and interpret it using the standard zones (above 2.99 safe; below 1.80 distress).